Curriculum Vitae

Personal Data

Pietro Benedusi, Ph.D.

Research fellow at the Euler Institute and scientific collaborator at the Istituto Ricerche Solari (IRSOL), Università della Svizzera italiana (USI), Lugano, Switzerland

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Research Interests

Numerical analysis, scientific computing, multigrid and time-parallel methods, high-performance computing, large-scale applications in radiative transfer and computational electrophysiology, scientific machine learning.

Research Positions

4/2024 – present	Research fellow at the Euler Institute and scientific collaborator at the "Istituto ricerche solari Aldo e Cele Daccò" (IRSOL), USI, Lugano. Lecturer at USI.
4/2022 – 4/2024	Postdoctoral researcher at the Simula Research Laboratory as a computational neuroscientist and HPC expert, Oslo.
6/2020 – 3/2022	Postdoctoral researcher at the Euler Institute, USI, Lugano.
8/2019 – 10/2019	Research fellow at the Berkeley Lab.
1/2014 – 9/2015	Student assistant at USI.

Education

10/2015 – 5/2020	PhD in Computational Science, USI, under the supervision of Prof. Rolf Krause.
9/2013 – 9/2015	Master's degree with summa cum laude in Applied Mathematics and Computational Science at USI joint with ETH Zurich.
9/2009 – 4/2013	Bachelor's degree in Physics at University of Pavia.

Selected Achievements and Activities

2026	PASC27 conference, Papers Program Committee Member (Computational Methods and Applied Mathematics).
11/2025	Italian habilitation as Associate Professor in Numerical Analysis (unanimous decision of the commission), ASN 23.

2025 PASC26 conference, Papers Program Committee Member (Computational Methods and Applied Mathematics).

2025 Offered a tenure-track professorship at UM6P, Morocco.

2025 Co-organized the mini-symposium "Advances in Iterative Solution Methods for Solving Coupled Problems" at COUPLED PROBLEMS 2025, Villasimius, Italy.

2025 Granted, as Co-PI, 1 million CPU hours for the project "Towards extreme-scale simulations of polarized radiative transfer for solar physics" at the Barcelona Supercomputing Center.

2024 Granted, as Co-PI, 2 million CPU hours for the project "Modeling the polarization of the chromospheric Ca I 4227 Å line in 3D models of the solar atmosphere with partial frequency redistribution" at the Barcelona Supercomputing Center.

2023 Organized the mini-symposium "High-performance computing tools for mixed dimensional problems" at SIAM CSE23, Amsterdam, Netherlands.

2022 Organized the mini-symposium "Scalable multilevel and multi-fidelity strategies for radiative transfer problems" at the International Multigrid Conference, Lugano, Switzerland.

2022 Granted a development project for computing resources at the Swiss National Supercomputing Centre (CSCS), in the context of solar simulations.

2019 Winner of the SIAM Travel Award grant to attend the CSE19 conference held in Spokane, USA.

2017 Organization of the summer school at USI: "Generalized locally Toeplitz sequences: a spectral analysis tool for discretized differential equations", Lugano, Switzerland.

2015 Recipient of the "USIimpresa" award and grant for best student in the Faculty of Informatics.

Teaching and Supervision

2025 – present Co-supervision of Melanie Tonarelli, PhD candidate at USI.

2025 Lecturer of the Master's course "Scientific Machine Learning" at USI (graded 9.4/10 by students).

2025 Lecturer of the PhD course "Advanced Tools in Computational Physiology" at the University of Insubria.

2024 Lecturer at the doctoral Summer School in Computational Physiology at the Simula Research Laboratory and for the MICROCARD

	workshop. Title: "Cell-by-cell modeling tutorial".
2023 – present	Co-supervision of four Master's theses in Computational Science (USI and Simula).
2023	Lecturer at the doctoral Summer School in Computational Physiology at the Simula Research Laboratory (joint event between the University of California San Diego and the University of Oslo). Title: "The EMI Model: applications in cardiac tissue and neuroscience and code-based exercises".
2022 – 2026	Co-supervision of Halvor Herlyng, PhD candidate in the Department of Scientific Computing at Simula.
2020 – 2023	Co-supervision of Simone Riva, PhD in Computational Science at USI (awarded August 2023).
2017 – 2020	Teaching assistant at USI for: Functional and Numerical Analysis, Theory of Computation, Fast Solvers, Introduction to PDEs, Linear Algebra, and Introduction to Computational Science.

Conference Presentations and Invited Talks

4/2026	Speaker at the Swiss Numerics Day, Zurich, Switzerland.
9/2025	Invited speaker at the 11th Solar Polarization Workshop, Prague, Czech Republic.
5/2025	Speaker at the XI International Conference on Coupled Problems in Science and Engineering, Villasimius, Italy.
2/2025	Speaker at the International Multigrid Conference, KAUST, Saudi Arabia.
2/2025	Speaker at the "Due Giorni di Algebra Lineare Numerica" conference, Pisa, Italy.
12/2024	Invited speaker at the Math. & Decisions conference, Rabat, Morocco.
6/2024	Speaker at the FEniCS conference, Oslo, Norway.
6/2024	Speaker at the ECCOMAS conference, Lisbon, Portugal.
1/2024	Speaker at DD28, KAUST, Saudi Arabia.
11/2023	Speaker at SfN, Washington, USA.
8/2023	Speaker at ICIAM, Tokyo, Japan.
7/2023	Speaker at the MICROCARD workshop, Strasbourg, France.
11/2022	Speaker at SPW10, Kyoto, Japan.

9/2022	Speaker at GIMC SIMAI YOUNG 2022, Pavia, Italy.
8/2022	Speaker at IMG2022, Lugano, Switzerland.
7/2022	Speaker at the 11th Parallel-in-Time Workshop, CIRM, France.
6/2022	Speaker at the ECCOMAS conference, Oslo, Norway.
2/2022	Speaker at the SIAM PP22 conference, held online.
9/2021	Speaker at the SIMAI 2021 conference, Parma, Italy.
8/2021	Speaker at the USNCCM16 conference and at the 10th Parallel-in-Time Workshop, both held online.
3/2021	Speaker at the SIAM CSE21 conference, held online.
12/2020	Speaker at the FOMICS-DADSI winter school on cardiac simulations, held online.
6/2020	Speaker at the 9th Parallel-in-Time Workshop, held online.
5/2019	Speaker at the 8th Parallel-in-Time Workshop, Bielefeld, Germany.
2/2019	Speaker at the SIAM CSE19 conference, Spokane, USA.
7/2018	Speaker at the DD25 conference, St. John's, Canada.
4/2018	Poster at the Swiss Numerics Day, ETH Zurich, Switzerland.
10/2017	Speaker at the 6th Parallel-in-Time Workshop, Ascona, Switzerland.
6/2017	Speaker at the LSSC17 conference, Sozopol, Bulgaria.
3/2017	Speaker at the 18th Copper Mountain Conference on Multigrid Methods, Colorado, USA.
2/2017	Speaker at the DD24 conference, Svalbard, Norway.
6/2016	Speaker at the MAFELAP conference, London, UK.
6/2016	Speaker at the ECCOMAS conference, Crete, Greece.
3/2016	Speaker at the SIAM PP16 conference, Paris, France.
12/2015	Speaker at the CBC Workshop on Dynamic Adaptivity, Simula, Oslo, Norway.

Computational Tools

C/C++, MATLAB, Python. Good knowledge of protocols for parallel computing (MPI and OpenMP) and HPC frameworks (e.g., Slurm). Experience with finite element libraries (LibMesh and FEniCS), meshing tools, numerical packages (PETSc, Trilinos), and PyTorch.

Publications

[1] S. Riva, G. Janett, F. Zeuner, L. Belluzzi, F. Riva, and P. Benedusi, "Validity of the complete frequency redistribution limit for modeling scattering polarization in the photospheric Sr I 4607 Å line," *Astronomy & Astrophysics*, vol. 714, p. A494, 2026.

[2] P. Benedusi, S. Riva, T. del Pino Alemán, G. Janett, F. Riva, J. Štěpán, R. Krause, J. Trujillo Bueno, and L. Belluzzi, "3D radiative transfer modeling of scattering polarization with partial frequency redistribution. I. Verification and disk-center results for the solar Ca I line at 4227 Å," *Astronomy & Astrophysics*, vol. 715, p. A83, 2026.

[3] P. Benedusi, S. Riva, L. Belluzzi, and S. Serra-Capizzano, "Analysis of eigenvalue clustering leads to optimal scaling in numerical radiative transfer," *Numerical Linear Algebra with Applications*, vol. 33, no. 5, p. e70118, 2026.

[4] P. Benedusi, P. Ferrari, M. Causemann, and S. Serra-Capizzano, "Dense cell-by-cell systems of PDEs: approximation, spectral analysis, and preconditioning," *BIT Numerical Mathematics*, vol. 66, p. 53, 2026.

[5] M. Tonarelli, S. Riva, P. Benedusi, F. Ferrandi, and R. Krause, "Adaptive multidimensional quadrature on multi-GPU systems," in *Proceedings of the 29th International Conference on Domain Decomposition Methods (DD29)*, 2026.

[6] F. Riva, G. Janett, L. Belluzzi, T. del Pino Alemán, E. Alsina Ballester, J. Trujillo Bueno, P. Benedusi, S. Riva, and R. Krause, "A numerical approach for modelling the polarisation signals of strong resonance lines with partial frequency redistribution. Applications to two-term atoms and plane-parallel atmospheres," *Astronomy & Astrophysics*, vol. 699, p. A233, 2025.

[7] A. J. Ellingsrud, P. Benedusi, and M. Kuchta, "A splitting, discontinuous Galerkin solver for the cell-by-cell electroneutral Nernst–Planck framework," *SIAM Journal on Scientific Computing*, vol. 47, no. 2, pp. B477–B504, 2025.

[8] A. Gatti, P. Benedusi, and S. Pezzuto, "Cell-by-cell modelling of electrical signaling in myelinated axons," in *Proceedings of the 28th International Conference on Domain Decomposition Methods (DD28)*, 2024.

[9] P. Benedusi, A. J. Ellingsrud, H. Herlyng, and M. E. Rognes, "Scalable approximation and solvers for ionic electrodiffusion in cellular geometries," *SIAM Journal on Scientific Computing*, vol. 46, no. 5, pp. B725–B751, 2024.

[10] L. Belluzzi, S. Riva, G. Janett, N. Guerreiro, F. Riva, P. Benedusi, T. del Pino Alemán, E. Alsina Ballester, J. Trujillo Bueno, and J. Štěpán, "Accurate PRD modeling of the forward-scattering Hanle effect in the chromospheric Ca I 4227 Å line," *Astronomy & Astrophysics*, vol. 691, p. A278, 2024.

[11] G. Janett, P. Benedusi, and F. Riva, "Numerical solutions to linear transfer problems of polarized radiation. IV. Efficient preconditioning in a physics-based framework," *Astronomy & Astrophysics*, vol. 682, p. A68, 2024.

[12] P. Benedusi, P. Ferrari, M. E. Rognes, and S. Serra-Capizzano, "Modeling excitable cells with the EMI equations: spectral analysis and iterative solution strategy," *Journal of Scientific Computing*, vol. 98, no. 3, p. 58, 2024.

[13] N. Guerreiro, G. Janett, S. Riva, P. Benedusi, and L. Belluzzi, "Modeling the scattering polarization in the solar Ca I 4227 Å line with angle-dependent PRD effects and bulk velocities," *Astronomy & Astrophysics*, vol. 683, p. A207, 2024.

[14] S. Riva, N. Guerreiro, G. Janett, D. Rossinelli, P. Benedusi, R. Krause, and L. Belluzzi, "Assessment of the CRD approximation for the observer's frame RIII redistribution matrix," *Astronomy & Astrophysics*, vol. 679, p. A87, 2023.

[15] P. Benedusi, S. Riva, P. Zulian, J. Štěpán, L. Belluzzi, and R. Krause, "Scalable matrix-free solver for 3D transfer of polarized radiation in stellar atmospheres," *Journal of Computational Physics*, vol. 479, p. 112013, 2023.

[16] P. Benedusi, G. Janett, S. Riva, R. Krause, and L. Belluzzi, "Numerical solutions to linear transfer problems of polarized radiation. III. Parallel preconditioned Krylov solver tailored for modeling PRD effects," *Astronomy & Astrophysics*, vol. 664, p. A197, 2022.

[17] P. Benedusi, G. Janett, L. Belluzzi, and R. Krause, "Numerical solutions to linear transfer problems of polarized radiation. II. Krylov methods and matrix-free implementation," *Astronomy & Astrophysics*, vol. 655, p. A88, 2021.

[18] G. Janett, P. Benedusi, L. Belluzzi, and R. Krause, "Numerical solutions to linear transfer problems of polarized radiation. I. Algebraic formulation and stationary iterative methods," *Astronomy & Astrophysics*, vol. 655, p. A87, 2021.

[19] P. Benedusi, M. L. Minion, and R. Krause, "An experimental comparison of a space-time multigrid method with PFASST for a reaction-diffusion problem," *Computers & Mathematics with Applications*, vol. 99, pp. 162–170, 2021.

[20] P. Benedusi, P. Ferrari, C. Garoni, R. Krause, and S. Serra-Capizzano, "Fast parallel solver for the space-time IgA-DG discretization of the diffusion equation," *Journal of Scientific Computing*, vol. 89, no. 1, pp. 1–21, 2021.

[21] S. Ben Bader, P. Benedusi, A. Quaglino, P. Zulian, and R. Krause, "Space-time multilevel Monte Carlo methods and their application to cardiac electrophysiology," *Journal of Computational Physics*, vol. 433, p. 110164, 2021.

[22] P. Benedusi, C. Garoni, R. Krause, X. Li, and S. Serra-Capizzano, "Space-time FE-DG discretization of the anisotropic diffusion equation in any dimension: the spectral symbol," *SIAM Journal on Matrix Analysis and Applications*, vol. 39, no. 3, pp. 1383–1420, 2018.

[23] A. Kreienbuehl, P. Benedusi, D. Ruprecht, and R. Krause, "Time-parallel gravitational collapse simulation," *Communications in Applied Mathematics and Computational Science*, vol. 12, no. 1, pp. 109–128, 2017.

[24] L. Grasedyck, C. Löbbert, G. Wittum, A. Nägel, V. Schulz, M. Siebenborn, R. Krause, P. Benedusi, U. Küster, and B. Dick, "Space and time parallel multigrid for optimization and uncertainty quantification in PDE simulations," in *Software for Exascale Computing – SPPEXA 2013–2015*, pp. 507–523, Springer, 2016.

[25] P. Benedusi, D. Hupp, P. Arbenz, and R. Krause, "A parallel multigrid solver for time-periodic incompressible Navier–Stokes equations in 3D," in *Numerical Mathematics and Advanced Applications – ENUMATH 2015*, pp. 265–273, Springer, 2016.

[26] P. Benedusi, *Parallel space-time multilevel methods with application to electrophysiology.*
PhD thesis, Università della Svizzera italiana, 2020.

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